

Web Data Integration  
Project Outline

**Pharmaceutical Drug Integration: Linking  
US and EU Approval Data**

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## 1 Brief description of use case

This project aims to build a unified view of approved pharmaceutical drugs in the United States and the European Union, enabling researchers and analysts to query regulatory status, approval history, and product attributes across both markets from a single integrated data set. To achieve this, we combine data from three complementary sources. Drugs@FDA is the FDA’s official database of US-approved drugs. EMA Medicines is the European Medicines Agency’s register of EU-authorised medicines. The openFDA NDC Directory is a supplementary US drug listing with broader product coverage. The central entity is the active substance, the pharmacologically active ingredient in a drug product. Active substances serve as a natural integration key because they persist across different brand names, manufacturers, regulatory jurisdictions, and formulations. The same substance can even carry different names across regions, as with paracetamol in the EU and acetaminophen in the US. Beyond regional synonyms, biosimilars carry manufacturer-specific suffixes that further complicate direct matching. Resolving these naming variations through identity resolution is therefore a central task of the project, ultimately producing one clean record per active substance with consolidated attributes from all three sources.

## 2 How the datasets fulfill the requirements

### 2.1 Schema and basic profile of each dataset

Table 1 summarises the three input datasets and lists the most relevant attributes, including those with high missingness.

#### Drugs@FDA

Drugs@FDA contains 51,864 product records across 29,377 applications. The product table contains only 8,388 distinct drug names and 3,355 distinct active-ingredient strings, showing substantial reuse across products. Consequently, DrugName + ActiveIngredient provides a more appropriate starting point for entity linkage than DrugName alone. The dataset is strongly imbalanced. Among applications, 78.3% are ANDAs, 20.1% NDAs, and 1.7% BLAs. Three attributes exceed 30% missingness: `ApplPublicNotes` (99.99%), `SubmissionsPublicNotes` (99.83%), and `ReviewPriority` (41.73%).

#### EMA Medicines

The EMA dataset contains 2,746 medicine records and 39 attributes, with no duplicate rows. It is predominantly composed of human medicines (85.6%). Medicine status is also imbalanced: 68.5% are currently authorised, 15.4% withdrawn, and 10.4% represent withdrawn applications. Fifty medicine names occur more than once, again showing that names alone do not uniquely identify entities. Twelve EMA attributes exceed 30% missingness, mostly due to structural reasons (e.g., veterinary attributes being inapplicable to human medicines).

Table 1: Overview of the input datasets

| Dataset       | Source (*) | Format   | # Ent.  | # Attr. | List of attributes (**)                                                                                                                                                                                                                                           |
|---------------|------------|----------|---------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Drugs@FDA     | U.S. FDA   | TXT/TSV  | 51,864  | 56      | ApplNo, ProductNo, Form, Strength, ReferenceDrug, DrugName, ActiveIngredient, ApplPublicNotes <sup>**</sup> , SponsorName, SubmissionType, SubmissionStatus, SubmissionsPublicNotes <sup>**</sup> , ReviewPriority <sup>**</sup> , MarketingStatusID, TECode, ... |
| EMA Medicines | EMA        | XLSX     | 2,746   | 39      | Name of medicine, EMA product number, Medicine status, Opinion status <sup>**</sup> , Active substance, Species (vet.) <sup>**</sup> , ATC code, Biosimilar, Generic, Orphan medicine, Opinion adopted date <sup>**</sup> , Revision number <sup>**</sup> , ...   |
| openFDA NDC   | U.S. FDA   | JSON→CSV | 138,209 | 10      | product NDC, brand name, generic name, active ingredients, manufacturer, dosage form, route, marketing start date, marketing category, application number                                                                                                         |

(\*) **Source:** Downloaded directly from the respective official provider websites.

(\*\*) **Missingness:** <sup>\*\*</sup> indicates >30% missing values. High missingness is mostly structural (e.g., optional notes, category-specific fields, or particular regulatory events).

### openFDA NDC Directory

The openFDA National Drug Code (NDC) Directory contains product listing information submitted to the FDA by drug labelers. It contains 138,209 product records and 7,682 distinct active ingredient entries. The listing includes both FDA-approved and unapproved products. Across the 10 selected columns, 113,976 of 1,382,090 values are missing (approx. 8.25%). Notable missingness includes `application_number` (27.56%), `manufacturer` (20.30%), and `route` (17.07%).

## 2.2 Integrated schema and overlap with input schemata

Because the entity is the active ingredient, the integrated schema aggregates product-level information from all three sources into one record per ingredient. Required transformations include splitting the Drugs@FDA Form field and the NDC active ingredients field, breaking down combination products, and normalising NDC application numbers. The integrated schema comprises 10 attributes (Table 2).

`activeIngredientName` is the primary matching key. `brandNames` and `manufacturers` occur in all three datasets and are modelled as lists (fusion by voting). `marketingStatus` assigns each ingredient to one of five classes, derived from source-specific status flags (fused by precedence: Active > Suspended > Pending > Discontinued > Not authorised). `usApprovalDate` and `euAuthorisationDate` describe distinct regulatory events and stay separate.

Table 2: Attribute Intersection with Integrated Schema

| Attribute name       | Attribute type       | Datasets in which the attribute is found |
|----------------------|----------------------|------------------------------------------|
| activeIngredientName | string               | Drugs@FDA, EMA Medicines, openFDA NDC    |
| brandNames           | list (string)        | Drugs@FDA, EMA Medicines, openFDA NDC    |
| manufacturers        | list (string)        | Drugs@FDA, EMA Medicines, openFDA NDC    |
| marketingStatus      | string (categorical) | Drugs@FDA, EMA Medicines, openFDA NDC    |
| dosageForms          | list (string)        | Drugs@FDA, openFDA NDC                   |
| routes               | list (string)        | Drugs@FDA, openFDA NDC                   |
| strengths            | list (string)        | Drugs@FDA, openFDA NDC                   |
| applicationNumbers   | list (string)        | Drugs@FDA, openFDA NDC                   |
| usApprovalDate       | date                 | Drugs@FDA                                |
| euAuthorisationDate  | date                 | EMA Medicines                            |

### 2.3 Why enough entities are likely contained in multiple datasets

The three datasets collectively cover a large and overlapping portion of the global pharmaceutical market. Both Drugs@FDA and EMA Medicines focus on formally approved drugs, meaning that any substance that has received regulatory clearance in both the US and EU will appear in both sources by definition. The openFDA NDC Directory adds further breadth as a comprehensive US product listing that includes both approved and marketed drugs.

Exact normalized **name** + **active substance** matching currently identifies 428 shared FDA-EMA entities, while normalized-name overlap reaches 752. Therefore, the required  $\geq 1,000$  shared entities have not yet been demonstrated. The third data set should, consequently, provide substantial overlap with the FDA or EMA and preferably contain active-substance information together with additional shared regulatory attributes.

As the larger of the two US sources, the openFDA NDC Directory is well positioned to fill the remaining gap. Since both FDA sources describe US-marketed products, their overlap is structurally guaranteed. Even a conservative 30% hit rate against Drugs@FDA entries would yield over 1,000 confirmed matches between the two US sources alone. Additionally, similarity-based matching will recover pairs that exact normalization misses, such as salt-form variants and biosimilar suffixes. Combining exact and approximate matching across all three sources, reaching 1,000 overlapping entities is a well-grounded expectation.